

Charge Response of One-Dimensional Electron Delocalization Relation Networks: Theoretical Prediction, Numerical Verification, and Boundary Revision Based on Relational Ontology

Guanghao Li¹

¹*Independent Researcher, Zhuangmu Town, Changfeng County, Hefei, China.
Member of Chinese Ethics Society, Member of Anhui Ethics Society.*

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We report a complete cycle of theoretical prediction, numerical testing, and self-correction. Starting from the Relational Ontology postulate, we define one-dimensional conjugated molecules as Electron Delocalization Relation Networks (EDRN). Under the assumption of a critical regime of classicality-annihilation phase transition, we derived a scaling exponent $\alpha = 1.75$ for the charge compressibility $\chi(0)$ as a function of chain length L . To test this, we performed Density Matrix Renormalization Group (DMRG) calculations on the one-dimensional half-filled Hubbard model at $U/t = 4$ for chains ranging from $L = 20$ to 200. A two-sector scheme based on particle-hole symmetry was designed to avoid the trivial null result of density-density correlation sums in the canonical ensemble. Eight independent data points consistently show that the charge gap converges to a constant $\Delta E \approx 1.35t$ and the compressibility converges to a constant $\chi \approx 0.74$, with a fitted scaling exponent $\alpha \approx -0.03$. The original prediction fails in the charge sector. Based on the experimental evidence, we diagnose that the system under test is in the Mott insulating phase, where charge degrees of freedom are frozen by the gap and do not contribute critical fluctuations. The anomalous dimension $\eta = 1/4$ and its scaling consequences should instead manifest in the spin sector. Strictly following the principles of Praxis-Intervention and Meta-Reflection, this work honestly reports the boundary of the theory under specific conditions and, through self-correction, points the direction to the true critical regime.

I. INTRODUCTION: FROM A PHILOSOPHICAL POSTULATE TO A TESTABLE PREDICTION

The reliability of any scientific theory does not rest on its aesthetic appeal or internal consistency but on its capacity to establish honest feedback loops with the real world. A theory must produce precise, testable predictions, and when data contradict those predictions, the theory must be revised. This is the core methodology of our work—the marriage of “Praxis-Intervention” and “Meta-Reflection” [1]. Cognition is intervention; theory must remain open to practice and be ready to amend any of its components at any time.

The starting point of the present theory is a basic philosophical postulate, the *Relational Ontology Postulate* [1]: The “real state” of any physical system is not determined by intrinsic properties of its constituent entities, but by the global configuration of the dynamic relation network formed among these entities. This relation network is the carrier of physical interactions; the observables of a system are manifestations of this network under specific interventions.

This is not a mere aphorism but a research program that can guide scientific investigation. It directly challenges the classical presupposition that “the world is made of entities with inherent attributes.” In condensed matter physics, particularly in strongly correlated electron systems, the “entity-first” mindset often fails. The behavior of an electron in a strongly correlated material is not dictated by its “inherent” mass and charge alone, but by the dynamic relation network it forms with the lattice and other electrons.

Based on this postulate, in a prior work [1] we constructed a theoretical framework that defines a one-dimensional conjugated molecule as an **Electron Delocalization Relation Network (EDRN)**. We hypothesized that such a network, at a specific quantum critical point—the critical regime of a classicality-annihilation phase transition (CAPT)—would exhibit universal scaling behavior. By assigning the criticality of the EDRN to the same universality class as the $S = 1/2$ Heisenberg antiferromagnetic chain, we introduced the anomalous dimension $\eta = 1/4$ and derived that the global response function $\chi(0)$ should follow a power-law scaling with chain length L : $\chi(0) \sim L^{2-\eta} = L^{1.75}$.

The central task of the present paper is to honestly report an independent numerical test of this theoretical prediction.

II. THE EDRN FRAMEWORK AND ORIGINAL PREDICTION

To make this paper self-contained, we briefly review the core structure of the EDRN theory. Detailed derivations can be found in Ref. 1.

Definition 1 (Electron Delocalization Relation Network, EDRN): For a conjugated molecule composed of N atoms, its EDRN is a weighted graph $G = (V, E, \omega)$. The nodes V represent atomic nuclei, the edges E represent effective electron delocalization interactions (orbital overlap integral $S_{ij} \geq 0.01$), and the edge weight $\omega_{ij} \in (0, 1]$ quantifies the strength of the “relation” established by electrons between nodes i and j .

Hypothesis 1 (Critical Universality Class): The low-energy physics of a one-dimensional EDRN at the CAPT critical point is dominated by spin-charge separation and belongs to the same universality class as the $S = 1/2$ Heisenberg antiferromagnetic chain [2, 3]. Therefore, its critical correlation function $G(r) \sim r^{-(d-2+\eta)}$ is characterized by the anomalous dimension $\eta = 1/4$ [4, 5].

Definition 2 (Global Collective Response Function): We focus on the network’s overall response to a local density perturbation $\hat{n}_0$. In the original theoretical construction, we defined

$$\chi(0) = \sum_{j=1}^L \langle n_0 n_j \rangle. \quad (1)$$

In the critical regime where the correlation length $\xi \sim L$, replacing the sum by an integral and inserting the critical correlation function yields the scaling relation

$$\chi(0) \sim \int_0^L G(r) dr \sim L^{2-\eta} = L^{1.75}. \quad (2)$$

This scaling relation is the core prediction of the theory. From it, we further linked, through an adiabatic toy model, to the scaling of the Diels-Alder reaction rate constant k [8, 9]:

$$k \sim \exp(-cL^{1.75}/k_B T). \quad (3)$$

We must now confront this prediction with rigorous numerical computation.

III. NUMERICAL METHODS AND COMPUTATIONAL PHILOSOPHY

The process of testing a prediction is itself an “intervention into practice.” The tool we choose (DMRG) and the observables we define constitute a specific cut into the theoretical field. We must therefore maintain transparency about the boundaries of our own intervention and strictly guard against the “paradox of instrumental rationality”—the temptation to fine-tune parameters during computation to “manufacture” the evidence the theory desires.

A. Model Selection and Experimental Design

We choose the one-dimensional single-band Hubbard model as an ideal platform to test the EDRN theory:

$$H = -t \sum_{\langle i,j \rangle, \sigma} (c_{i\sigma}^\dagger c_{j\sigma} + \text{h.c.}) + U \sum_i n_{i\uparrow} n_{i\downarrow}. \quad (4)$$

The parameters are set to $U/t = 4$, half-filling ($\langle n \rangle = 1$), and open boundary conditions. We compute ground-state properties for the chain-length sequence $L = 20, 40, 60, 80, 100, 120, 160, 200$. This choice of model and parameters directly corresponds to the “strongly correlated regime” hypothesized in the theory.

B. A Self-Corrected Measurement Scheme: From Three Sectors to Two

This investigation underwent a crucial self-correction in methodology, exemplifying the Meta-Reflection principle.

Stage 1 (Three-Sector Method). We initially planned to directly compute $\chi(0)$ as defined in Eq. (1). However, we immediately encountered a physical “silent dissonance”: in the canonical ensemble with fixed total particle number $N = L$, particle-number conservation forces the connected part of $\sum_j \langle n_0 n_j \rangle$ to be identically zero. This is a trivial mathematical fact that obscures the genuine physics. To obtain a nontrivial response, we turned to computing the **charge compressibility**, the derivative $\kappa = \partial \langle n \rangle / \partial \mu$. Numerically, it can be approximated by a finite difference: $\kappa \propto 1/(E(N+1) + E(N-1) - 2E(N))$. We wrote DMRG scripts to compute the ground-state energies in the three particle-number sectors $N = L - 1, L, L + 1$.

However, this scheme repeatedly stalled in calculations for $L = 60$ and $L = 120$, especially in the $N = L + 1$ sector. Diagnosis revealed that our initial-state construction for $N > L$ introduced a local doubly-occupied state (“full” state). Under strong repulsion $U/t = 4$, this initial guess lies too far from the true ground state (a delocalized charge carrier), causing extremely slow DMRG convergence. This is a trap of instrumental rationality: we were forcing an unreliable calculation to obtain the desired data points. Data indicated that compressibility values from this scheme were systematically too large and exhibited an anomalous trend.

Stage 2 (Two-Sector Method). Guided by Meta-Reflection, we abandoned the unreliable scheme. Exploiting the particle-hole symmetry near half-filling, we assume $E(N+1) - E(N) \approx E(N) - E(N-1)$. Consequently, the compressibility can be computed using only two sectors with excellent convergence properties:

$$\kappa \propto \frac{1}{E(L) - E(L-1)}. \quad (5)$$

Here $E(L)$ is the half-filled ground-state energy and $E(L-1)$ is the energy with one electron removed. The initial states for both sectors are Slater determinants without local double occupancy; they converge rapidly, yielding clean and reliable data. We recomputed the entire sequence $L = 20$ to 200 with identical parameter configurations. Each data point is an independent, transparent, and auditable verdict.

C. Technical Implementation and Audit Trail

We implement the DMRG algorithm using the TeNPy library [10]. In full adherence to the transparency principle, all calculations are performed on a local device without reliance on any cloud or remote server. All code, raw data, and parameter configurations are open-sourced

in the GitHub repository `edrn-dmrg-verification` [11]. The concrete computational parameters are: maximum bond dimension `chi_max=100`, truncation tolerance `svd_min=1e-10`, and maximum number of sweeps `max_sweeps=50`.

IV. RESULTS: WHEN DATA MEETS PREDICTION

The complete set of raw data obtained by the two-sector method is presented in Table I. The corresponding trends are plotted in Fig. ?? (schematic).

TABLE I. Charge compressibility data for the 1D half-filled Hubbard model ($U/t = 4$) at various chain lengths. $\Delta E = E(L) - E(L - 1)$ is the charge gap; $\chi = 1/\Delta E$ is the compressibility proxy. All energies are in units of t .

L	$E(L)$	$E(L - 1)$	ΔE	χ
20	-11.112183	-12.358507	1.246324	0.802360
40	-22.583571	-23.897547	1.313976	0.761049
60	-34.056974	-35.388735	1.331761	0.750885
80	-45.530881	-46.869781	1.338900	0.746882
100	-57.004967	-58.347281	1.342315	0.744982
120	-68.479122	-69.823216	1.344094	0.743996
160	-91.427500	-92.772989	1.345489	0.743224
200	-114.375928	-115.722211	1.346283	0.742786

The data analysis paints a clear, consistent physical picture:

1. **Charge gap converges to a constant.** The gap ΔE gradually increases from $1.246t$ at $L = 20$ and stabilizes at $1.346t$ for $L = 200$. Extrapolating to the thermodynamic limit ($L \rightarrow \infty$) gives a gap of approximately $1.35t$. A finite charge gap is the hallmark of a Mott insulator.
2. **Compressibility does not exhibit power-law divergence.** The compressibility χ decreases monotonically from 0.802 and rapidly saturates; for $L \geq 160$ it remains essentially constant at ≈ 0.743 .

A log-log linear fit $\log \chi \sim \alpha \log L$ to these eight points yields the scaling exponent

$$\alpha \approx -0.03 \pm 0.01. \quad (6)$$

This value is opposite in sign to the theoretical prediction $\alpha = 1.75$ and differs by nearly two orders of magnitude. The data unequivocally tell us: **In the half-filled Hubbard chain at $U/t = 4$, the charge compressibility does not show any critical power-law enhancement.**

V. DISCUSSION: WHERE IS THE THEORY? — AN EVIDENCE-BASED BOUNDARY REVISION

The data did not support our original prediction. Now, our Meta-Reflection principle demands that we answer: Why? Is the theory wrong? Where does it belong?

A. Root Cause: Mismatch between the Mott Gap and the Critical Regime

We failed to observe critical behavior because we were looking in the wrong place. Our theory assumed the system to be in the “CAPT critical regime,” but the real physical system—the half-filled 1D Hubbard model at $U/t = 4$ —is in the **Mott insulating phase**. The exact solution by Lieb and Wu [12] proved long ago that in one dimension at half-filling, any $U > 0$ opens a finite charge gap. Charge degrees of freedom are “frozen” and cannot participate in long-range critical fluctuations. The gap $\Delta E \approx 1.35t$ we computed is precisely the charge gap of the Mott insulator. In this phase, the charge compressibility is a constant, not a divergent power-law. Our DMRG data, in fact, successfully and accurately capture this well-known physical fact.

B. The True Location of the Theory: The Spin Sector

Our theoretical derivation was not entirely wrong. We correctly identified the universality class: the $S = 1/2$ Heisenberg antiferromagnetic chain with its anomalous dimension $\eta = 1/4$. This universality class describes not charge fluctuations, but **spin fluctuations**. Inside a Mott insulator, although charges are localized, the spin degree of freedom on each lattice site remains active. Through virtual hopping processes, an antiferromagnetic superexchange interaction $J \sim t^2/U$ arises between localized spins, and the low-energy effective theory is precisely the Heisenberg model [2, 3]. The spin-spin correlation function of the Heisenberg model indeed decays with the power law characterized by $\eta = 1/4$.

Therefore, our theory has not been “falsified.” It needs to be **re-localized**. The $\eta = 1/4$ and the scaling behavior we predicted find their true stage in the spin sector.

C. A Revised, Testable New Prediction

Based on this analysis, we propose a corrected, precise, and testable new prediction: For the same physical system (1D half-filled Hubbard model, $U/t = 4$), **the spin gap Δ_s must decay to zero as a power law with chain length, specifically $\Delta_s \sim 1/L$** . The energy difference between the spin triplet and spin singlet ground states should be gapless in this model, providing direct

evidence that the system is in a spin-critical regime. Unlike the charge response that is “silenced” by the Mott gap, the critical vanishing of the spin gap is the genuine, observable scaling signal of the $\eta = 1/4$ universality class.

VI. CONCLUSION

This work completes a full cycle from a philosophical postulate to a numerical test.

We started from a grand postulate, derived an elegant scaling-law prediction ($\alpha = 1.75$). Then, using an independent, honest, and self-critical DMRG scheme, we obtained eight clean data points for $L = 20$ to 200. The data showed that the original prediction fails in the charge sector.

Faced with this “negative” result, our Meta-Reflection principle guided us toward a theoretical revision rather than abandonment. We located the root of the conflict: our computation touched the boundary of the theory, not its core. The core of the theory—the relation network, the universality class, the critical scaling—remains intact; it has simply been correctly positioned in the spin

sector.

The conclusions are:

1. **Numerical finding:** Through independent DMRG calculations, we confirm that the 1D half-filled Hubbard model at $U/t = 4$ is a Mott insulator with a charge gap $\Delta E \approx 1.35t$; the charge compressibility shows no critical scaling.
2. **Theoretical revision:** The $\eta = 1/4$ critical behavior predicted by the EDRN framework does not apply to the charge sector but points to the spin sector.
3. **New prediction:** We propose a decisive new test: the spin gap of the same system should scale as $\Delta_s \sim 1/L$.

This is not merely a paper about a specific numerical value; it is a methodological demonstration. It proves that “Relational Ontology” is not a static philosophical assertion but a living cognitive engine capable of friction with the real world and of responding to feedback.

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